

Epilogue

This book was about the statistical analysis of *spatio-temporal* data. Throughout our exploration of this topic, we have repeatedly come back to a few key themes: dividing a complex problem into simpler components by thinking conditionally (i.e., hierarchical modeling); accounting for uncertainty in data, process, and parameters; the use of scientific principles and findings to motivate each level of a hierarchical model; the dynamical specification of spatio-temporal models; and data/model/computing compromises. Perhaps the best way to summarize this book is that it provides a systematic approach for quantification of uncertainty in spatio-temporal data in a way that scientific information can be easily incorporated in support of inference.

Admittedly, our coverage of this topic has been far from complete. Some topics were covered in less depth than we would have liked (e.g., descriptive spatio-temporal models, multivariate spatio-temporal processes, nonlinear dynamical models, dynamical design, sequential Monte Carlo); some things were hardly covered (e.g., change-point detection, spatio-temporal point processes, dynamical modeling of objects, spatio-temporal field comparison); and some things were not considered at all (e.g., model selection, extremes). In the end, we made decisions and choices to emphasize certain topics for myriad reasons. One of them was that this is a big book that simply could not get any bigger; time and space were limited!

Ultimately, we chose topics that we thought were fundamental and some topics that were, in some sense, “dear to our hearts.” We do feel that this book represents a fairly accurate snapshot of (a projection of) the state of statistical analysis of spatio-temporal data *circa* 2010. But this is a rapidly evolving field, and it is notoriously hard to hit a moving target. We have attempted to be scholarly, and we have tried to look beyond our own discipline for important contributions. We have almost certainly fallen short and unintentionally missed some useful and important papers. We offer our apologies for these oversights.

At the end of a long book like this, it is tempting to say something about the *future* of the field. As statisticians, we recognize that it is dangerous to extrapolate, but the opportunity to forecast is too great to resist.

We are certain that computational issues are always going to be linked closely with spatio-temporal statistical analysis. For most problems, the datasets are large, and they keep getting larger! In addition, as datasets grow, model complexity grows, and consequently there is a need for ever-increasing computing power and/or algorithms for efficient estimation and inference. Of course, there will always be trade-offs between what one would like a model to be and what the available tools will allow it to be, suggesting a modeling strategy of “dialing back” from what one would like to do, to what one is able to do. This is a “progressive” strategy that we expect will offer continual improvement of the state-of-the-art.

As for the field itself, we believe that we have only just begun to realize the potential of modeling complex spatio-temporal systems. Nonlinear interactions across many processes define the world in which we live. To address important scientific questions in the presence of uncertainty, we shall need to develop spatio-temporal statistical methodologies that can efficiently accommodate multivariate interactions in nonlinear and nonstationary environments. One promising avenue of research is the further development of individual-based (or, agent-based) models. We also believe that there will be increased use of a variety of deterministic “computer” models, and we see our topic playing a role in making such models more stochastic and more responsive to data. In addition, we believe there will be increased data on non-Euclidean spaces (such as networks, social or otherwise), that change with time. This will come as a result of our society becoming increasingly “connected.” In some sense, cyber networks, in addition to being non-Euclidean, also tend to change our perception of time. That is, information may travel much faster than it would have in the past, resulting in processes with vastly different “clock speeds” that interact in nonstationary ways.

Finally, we believe that spatio-temporal statistics has a large, and mainly untapped role to play in public policy. That is, as the questions we are faced with (e.g., global climate change, water resources, hunger, economic stability) become more critical to society in general, we believe strongly that honest quantification and accounting of uncertainty must play a bigger role in decision making, which is the apex of the knowledge pyramid described at the beginning of the book. Here, a summary based on the posterior mean and the posterior variance will not be enough. Development of multivariate loss functions (or, better, utility functions that involve cost components) are needed to quantify policy questions. Many of these issues are indeed spatio-temporal, and it is important that we discover better ways to incorporate our understanding of data, processes, and parameters into a decision-support environment. Importantly, it should be acknowledged that this is not just a matter of scientific and statistical discovery, since there is increasingly less distinction between policy and politics. Nevertheless, we can and should continue to provide information

and, better yet, knowledge, so that uncertainties and data/model issues are presented completely and openly. In this way, we shall do our part in contributing to “Science in the service of Society.”

This book is finished, and now we can see what wasn’t obvious before . . .

We shall not cease from exploration
And the end of all our exploring
Will be to arrive where we started
And know the place for the first time
(T. S. Eliot)

